HealthHub Systems

Software Development Company

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I. Executive Summary

The objective of this project was to develop a comprehensive database system to enhance the operational efficiency of health insurance administration, spanning subscription to claims processing and other ancillary activities. In pursuit of this goal, the team paid attention to design a database system which eliminates redundancies and increases efficiency. Our initial step involved conducting extensive research to thoroughly understand the various processes involved in health insurance management. Drawing on our collective experiences and the insights gained from individual research, each team member made significant contributions to the project.

Adhering to a structured methodology, we progressed through stages of conceptualization, logical modeling, physical modeling, database implementation, system analysis, and application development. Advanced normalization techniques were employed to ensure the database system was both efficient and effective. Our efforts culminated in the creation of HealthHub Systems.

HealthHub Systems is engineered to manage critical data related to subscribers and their dependents, insurance plans, claims, and provider information, among other elements. This capability enables healthcare insurance providers to access and analyze relevant business metrics. In subsequent sections, we present scenarios that demonstrate the capabilities of our system, providing insights into important business indices.

II. Introduction

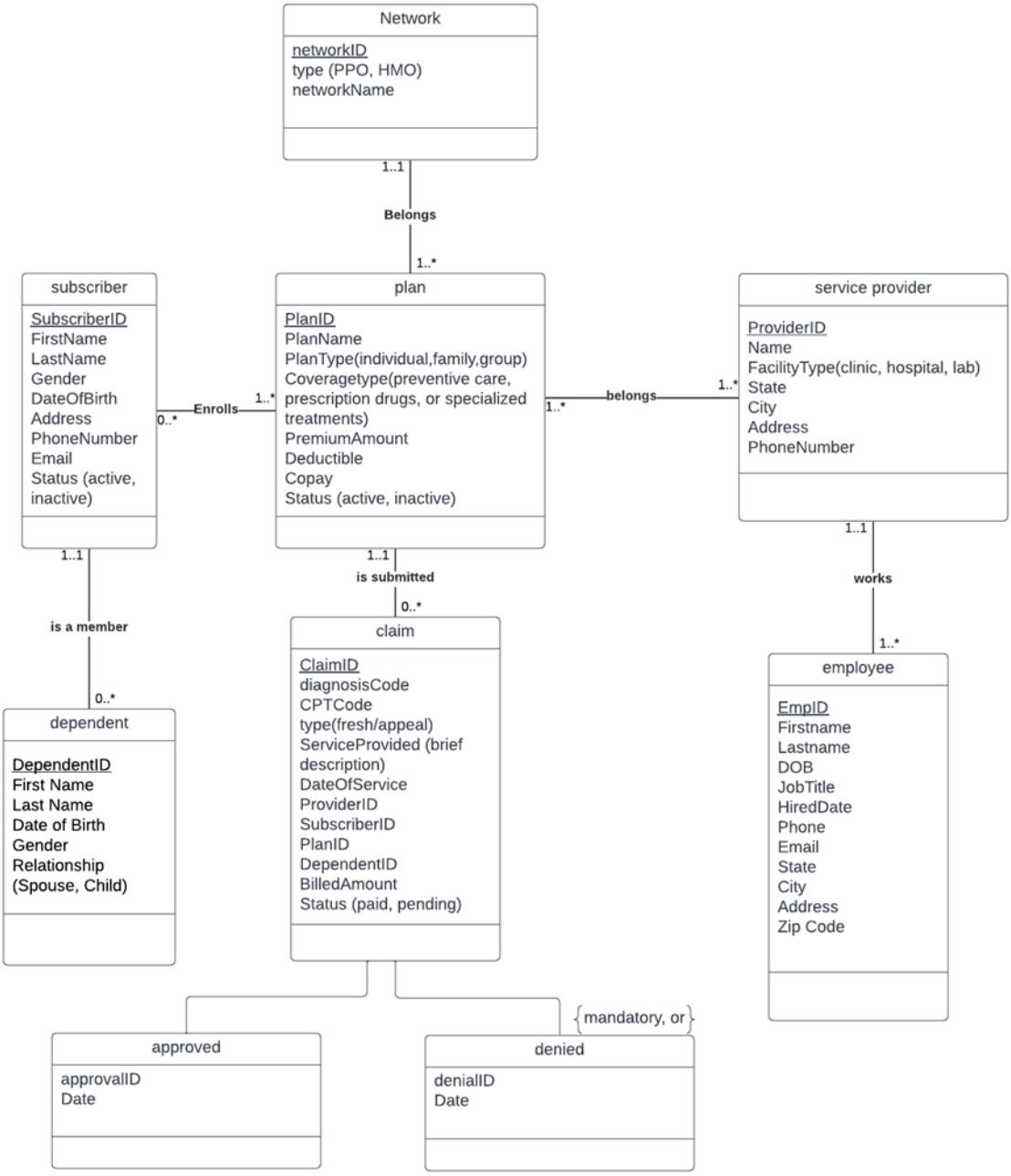
HealthHub Systems is a software development company specializing in facilitating the efficiency and effectiveness of interactions within the healthcare insurance industry. Recognizing the challenges in managing interactions between health insurance providers, their subscribers, and service providers, HealthHub Systems aims to revamp these processes by developing a comprehensive database system. With our system, subscribers can easily enroll in specific plans, manage dependents, and service providers can submit claims based on services received. We will also aim to solve the problem of streamlining insurance to ensure quality customer service. Our database will facilitate claim review processes and display payment or pending status, providing efficiency and overall satisfaction in the healthcare insurance industry.

Information Needs

To effectively solve the presented problem, we collected information on:

- Subscriber details: names, addresses, and contact details.
- Dependent's information: relationship to insurance holder, and coverage details.
- Plan Details: plan names, coverage types, and associated costs and benefits.
- Information on Service Providers: hospitals, clinics, or healthcare professionals participating in the networks such as their names, specialties, and contact information.
- Claim data: claim IDs, service dates, and billed amounts.
- Verification and Review information: reviewer (employee IDs), date, outcomes (approved or pending, not approved).
- Network information: doctor name, tax id, facilities, location.

III. ERD Model



Relationship Sentences

Network- Plan:

One Network has one or many Plans : 1..*

One Plan belongs to one and only one Network: 1..1

Plan- Service Provider:

One Plan is offered by one or many Service Providers: 1..*

One Service Provider offers one or many Plans : 1..*

Service Provider- Employee:

One Service Provider has one or many employees: 1..*

One employee works for one and only one Service Provider : 1..1

Plan- Subscriber:

One Plan has zero or many subscribers: 0..*

One Subscriber subscribes to one or many plans: 1..*

Subscriber- Dependent(s):

One Subscriber can have zero or many dependents: 0..*

One Dependent can belong to one and only one subscriber: 1..1

Plan- Claim

One Plan has zero or many claims: 0..*

One claim belongs to one and only one plan: 1..1

Claim- Approved or Denied

Approved

Mandatory, OR

Denied

Mandatory, OR

IV. RDM

1	Network (networkID, type, networkName)
2	Plan (Plan ID, PlanName, PlanType, Coveragetype, PremiumAmount, Deductible, Copay, Status, NetworkID (fk))
3	Subscriber (SubscriberID, FirstName, LastName, Gender, DateOfBirth, Address, PhoneNumber, Email, Status)
4	Plan_Subsriber (Plan ID (fk), SubscriberID (fk))
5	Dependent (DependentID, First Name, Last Name, Date of Birth, Gender, Relationship, SubscriberID (fk))
6	Service Provider (ProviderID, Name, FacilityType, State, City, Address, PhoneNumber)
7	Plan_Service Provider (Plan ID (fk), Provider ID (fk))
8	Employee (EmpID, Firstname, Lastname, DOB, JobTitle, HiredDate, Phone, Email, State, City, Address, Zip Code, ProviderID (fk))
9	Claim_Approved (ClaimID, diagnosisCode, CPTCode, type, ServiceProvided, DateOfService, ProviderID, SubscriberID, PlanID, DependentID, BilledAmount, Status, ApprovalID, Date, PlanID)
10	Claim_Denied (ClaimID, diagnosisCode, CPTCode, type, ServiceProvided, DateOfService, ProviderID, SubscriberID, PlanID, DependentID, BilledAmount, Status, DenialID, Date, PlanID)

V. Normalization

Relation 1

Step 1: Primary Key? Yes: NetworkID in 1NF

Step 2: Functional Dependencies:

FD1	Network	----->	(networkID, type, networkName)
FD2	NetworkID	----->	(Type, NetworkName)

Do we have partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

Relation 2

Step 1: Primary Key? Yes: Plan ID 1NF

Step 2: Functional Dependencies:

FD1	Plan	----->	(Plan ID, PlanName, PlanType, Coveragetype, PremiumAmount, Deductible, Copay, Status, NetworkID)
FD2	PlanID	----->	(PlanName, PlanType, Coveragetype, PremiumAmount, Deductible, Copay, Status, NetworkID)

Do we have partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

Relation 3

Step 1: Primary Key? Yes: SubscriberID 1NF

Step 2: Functional Dependencies:

FD1	Subscriber	----->	(SubscriberID, FirstName, LastName, Gender, DateOfBirth, Address, PhoneNumber, Email, Status)
FD2	SubscriberID	----->	(FirstName, LastName, Gender, DateOfBirth, Address, PhoneNumber, Email, Status)

Do we have partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

Relation 4

Step 1: Primary Key? Yes: Plan ID, SubscriberID 1NF

Step 2: Functional Dependencies:

FD1	Plan_Subsriber	----->	(Plan ID, SubscriberID)
FD2	SubscriberID	----->	(PlanID)

Do we have partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

Relation 5

Step 1: Primary Key? Yes: DependentID 1NF

Step 2: Functional Dependencies:

FD1	Dependent	----->	(DependentID, First Name, Last Name, Date of Birth, Gender, Relationship, SubscriberID)
FD2	DependentID	----->	(First Name, Last Name, Date of Birth, Gender, Relationship, SubscriberID)

Do we have partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

Relation 6

Step 1: Primary Key? Yes: ProviderID 1NF

Step 2: Functional Dependencies:

FD1	Service Provider	----->	(ProviderID, Name, FacilityType, State, City, Address, PhoneNumber)
FD2	ProviderID	----->	(Name, FacilityType, State, City, Address, PhoneNumber)

Do we have a partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

Relation 7

Step 1: Primary Key? Yes: PlanID , ProviderID 1NF

Step 2: Functional Dependencies:

FD1	Plan_Service Provider	----->	(PlanID , ProviderID)
FD2	Plan ID	----->	(ProviderID)

Do we have partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

Relation 8

Step 1: Primary Key? Yes: EmpID 1NF

Step 2: Functional Dependencies:

FD1	Employee	----->	(EmpID, Firstname, Lastname, DOB, JobTitle, HiredDate, Phone, Email, State, City, Address, Zip Code, ProviderID)
FD2	EmpID	----->	(Firstname, Lastname, DOB, JobTitle, HiredDate, Phone, Email, State, City, Address, Zip Code, ProviderID)

Do we have partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

Relation 9

Step 1: Primary Key? Yes: ClaimID 1NF

Step 2: Functional Dependencies:

FD1	Claim_Approved	-->	(ClaimID, diagnosisCode, CPTCode, type, ServiceProvided, DateOfService, ProviderID, SubscriberID, DependentID, BilledAmount, Status, ApprovalID, Date, PlanID)
FD2	ClaimID	-->	(diagnosisCode, CPTCode, type, ServiceProvided, DateOfService, ProviderID, SubscriberID, PlanID, DependentID, BilledAmount, Status, ApprovalID, Date, PlanID)

Do we have partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

Relation 10

Step 1: Primary Key? Yes: ClaimID 1NF

Step 2: Functional Dependencies:

FD1	Claim_Denied	-->	(ClaimID, diagnosisCode, CPTCode, type, ServiceProvided, DateOfService, ProviderID, SubscriberID, PlanID,DependentID, BilledAmount,Status, DenialID, Date, PlanID)
FD2	ClaimID	-->	(diagnosisCode, CPTCode, type, ServiceProvided, DateOfService, ProviderID, SubscriberID, PlanID, DependentID, BilledAmount, Status, DenialID, Date, PlanID)

Do we have partial-key dependency, meaning a part of (primary) key that determines a non-key attribute(s)? Recall this happens only in a composite key.

R1	Checked, no more, in 2NF

Step 3: Do we have transitive dependency, meaning do we have a Non key attribute that determines one non-key attribute?

R1	Checked, no more, in 3NF

Step 4: Do we have a non-key attribute that determines part of the key? No

Checked, no more, in BCNF

Final Relations: R1

VI. SQL DDL

Note: Built in MS SQL Server

CREATE DATABASE HEALTHHUBSYSTEMS
use HEALTHHUBSYSTEMS

```
CREATE TABLE network
(networkID int not null,
type varchar(50),
networkName varchar(50)
constraint pk_network primary key (networkID)
);
```

```
CREATE TABLE [plan]
(planID int not null,
planName varchar(50),
planType varchar(50),
coverageType varchar(50),
premiumAmount float,
deductible float,
copay float,
[status] varchar(50),
networkID int,
constraint pk_plan primary key (planID),
constraint fk_plan_network foreign key(networkID)
references network(networkID)
);
```

```
CREATE TABLE subscriber
(subscriberID int not null,
firstName varchar(50),
lastName varchar (50),
gender varchar(10),
dateOfBirth date,
[address] varchar (100),
phoneNumber varchar(13),
email varchar(50),
[status] varchar(50),
constraint pk_subscriber primary key (subscriberID));
```

```
CREATE TABLE plan_subscriber
(planID int not null,
subscriberID int not null,
constraint pk_plan_subscriber primary key (planID,
subscriberID))
```

```
CREATE TABLE [dependent]
(dependentID int not null,
firstName varchar(50),
lastName varchar(50),
dateOfBirth date,
gender varchar(10),
relationship varchar(10),
subscriberID int,
constraint pk_dependent primary key(dependentID),
constraint fk_dependent_subscriber foreign key(subscriberID)
references subscriber(subscriberID)
)
```

```
CREATE TABLE [provider]
(providerID int not null,
[name] varchar(50),
facilityType varchar(50),
[state] varchar(20),
city varchar (50),
[address] varchar(100),
phoneNumber varchar(13),
constraint pk_provider primary key(providerID));
```

```
CREATE TABLE plan_provider
(planID int not null,
providerID int not null,
constraint pk_plan_provider primary key(planID, providerID))
```

```
CREATE TABLE employee
(empID int not null,
firstName varchar(50),
lastName varchar(50),
dateOfBirth date,
jobTitle varchar(50),
hiredDate date,
phoneNumber varchar(13),
```

```
email varchar(50),
[state] varchar(20),
city varchar(50),
[address] varchar(100),
Zip_code varchar(5),
providerID int
constraint pk_employee primary key(empID),
constraint fk_employee_provider foreign key(providerID)
references [provider] (providerID))
```

```
CREATE TABLE claim_approved
(claimID int not null,
diagnosisCode varchar(10),
CPTCode varchar(10),
[type] varchar(20),
serviceProvided varchar(200),
dateOfService date,
providerID int,
subscriberID int,
planID int,
dependentID int,
billedAmount float,
[status] varchar(50),
approvalID varchar(5),
submissionDate date,
 constraint pk_claim_approved primary key(claimID),
 constraint fk_claim_approved_provider foreign key (providerID)
 references [provider] (providerID),
 constraint fk_claim_approved_subscriber foreign
key(subscriberID)
 references subscriber(subscriberID),
 constraint fk_claim_approved_plan foreign key (planID)
 references [plan] (planID),
 constraint fk_claim_approved_dependent foreign key(dependentID)
 references [dependent](dependentID));
```

```
CREATE TABLE claim_denied
(claimID int not null,
diagnosisCode varchar(10),
CPTCode varchar(10),
```

```
[type] varchar(20),
serviceProvided varchar(200),
dateOfService date,
providerID int,
subscriberID int,
planID int,
dependentID int,
billedAmount float,
[status] varchar(50),
denialID varchar(5),
submissionDate date,
 constraint pk_claim_denied primary key(claimID),
 constraint fk_claim_denied_provider foreign key (providerID)
 references [provider] (providerID),
 constraint fk_claim_denied_subscriber foreign key(subscriberID)
 references subscriber(subscriberID),
 constraint fk_claim_denied_plan foreign key (planID)
 references [plan] (planID),
 constraint fk_claim_denied_dependent foreign key(dependentID)
 references [dependent](dependentID));
```

Insert Statements

```
insert into network values
 (1001,'HMO','Network C');
```

	networkID	type	networkName
1	1001	HMO	Network C
2	1002	POS	Network B
3	1003	PPO	Network C
4	1004	HMO	Network C
5	1005	EPO	Network A
6	1006	PPO	Network D
7	1007	POS	Network A
8	1008	HMO	Network A
9	1009	PPO	Network A
10	1010	EPO	Network B

```
insert into [plan] values
(3001,'Plan A', 'HMO', 'Platinum',7366,522,72,'Inactive', 1004);
```

	planID	planName	planType	coverageType	premiumAmount	deductible	copay	status	networkID
1	3001	Plan A	HMO	Platinum	7366	522	72	Inactive	1004
2	3002	Plan C	EPO	Gold	11252	1232	53	Active	1007
3	3003	Plan E	POS	Bronze	16321	823	50	Inactive	1002
4	3004	Plan E	POS	Platinum	14720	563	75	Inactive	1007
5	3005	Plan B	EPO	Bronze	11604	1534	71	Inactive	1004
6	3006	Plan E	POS	Gold	15848	283	70	Inactive	1006
7	3007	Plan D	EPO	Silver	15735	982	50	Active	1004
8	3008	Plan C	POS	Gold	12051	560	67	Inactive	1010
9	3009	Plan E	EPO	Gold	6098	829	51	Inactive	1008
10	3010	Plan D	HMO	Silver	10991	366	41	Active	1001

```
insert into plan_provider values
(3005,    6007);
```

	planID	providerID
1	3001	6010
2	3002	6002
3	3002	6009
4	3004	6008
5	3005	6001
6	3005	6007
7	3007	6002
8	3009	6001
9	3009	6002
10	3009	6010

```
insert into subscriber values
(4001,'Wildon','Lisciardelli','Male',   '1987-07-12','2 Dapin
Parkway', '117-857-6767', 'wlisciardelli0@answers.com',
'active');
```

	subscriberID	firstName	lastName	gender	dateOfBirth	address	phoneNumber	email	status
1	4001	Wildon	Lisciardelli	Male	1987-07-12	2 Dapin Parkway	117-857-6767	wlisciardelli0@answers.com	active
2	4002	Gusty	Passby	Female	1969-10-06	45099 Veith Pass	439-335-2857	gpassby1@whitehouse.gov	inactive
3	4003	Marybeth	Macauley	Female	1992-03-04	70 Oak Center	887-584-2426	mmacauley2@umich.edu	active
4	4004	Mohandas	Meeny	Male	1960-12-31	74 3rd Parkway	899-439-9477	mmeeny3@nytimes.com	inactive
5	4005	Cozmo	Wafer	Male	1953-09-21	2369 Vernon Plaza	371-260-8825	cwafer4@fda.gov	active
6	4006	Addison	Kibard	Male	1976-06-16	170 Algoma Alley	785-932-6223	akibard5@dion.ne.jp	inactive
7	4007	Josie	Cordero	Male	1961-10-26	91260 Eliot Drive	457-167-5214	jcordero6@patch.com	inactive
8	4008	Thea	Beneix	Female	1961-04-17	716 Sugar Point	999-643-6283	tbeneix7@ft.com	active
9	4009	Franchot	Guilayn	Male	1996-06-12	3331 Lyons Hill	887-791-5994	fguilayn8@google.cn	inactive
10	4010	Pinchas	Dumbarton	Male	1998-09-03	248 Lukken Center	492-372-2996	pdumbarton9@last.fm	inactive

```
insert into [dependent] values
(5001,'Wendye','Colgan', '2020-04-22','Female','spouse', 4001);
```

	dependentID	firstName	lastName	dateOfBirth	gender	relationship	subscriberID
1	5001	Wendye	Colgan	2020-04-22	Female	spouse	4001
2	5002	Leif	Lacroutz	2016-05-06	Male	child	4002
3	5003	Jim	Dionisetto	2011-03-25	Male	spouse	4003
4	5004	Rafferty	Fortnon	2021-08-23	Male	child	4004
5	5005	Aubert	Jodlkowski	2011-03-27	Male	child	4005
6	5006	Clotilda	Storch	2019-05-08	Female	spouse	4006
7	5007	Conny	Blew	2009-04-25	Female	child	4007
8	5008	Curran	McGarvey	2016-09-15	Male	child	4008
9	5009	Arman	Ventum	2010-08-23	Male	child	4009
10	5010	Dieter	Alvin	2018-06-07	Male	child	4010

```
insert into plan_subscriber values
(3004,    4005);
```

	planID	subscriberID
1	3002	4007
2	3003	4004
3	3004	4005
4	3005	4003
5	3005	4004
6	3006	4004
7	3006	4005
8	3007	4007
9	3008	4001
10	3009	4004

```
insert into [provider] values
(6001, 'Provider CC', 'Hospital', 'Florida', 'Fort Lauderdale'
,'64 Nobel Crossing', '954-739-0363');
```

	providerID	name	facilityType	state	city	address	phoneNumber
1	6001	Provider CC	Hospital	Florida	Fort Lauderdale	64 Nobel Crossing	954-739-0363
2	6002	Provider BB	Specialist Service	Arizona	Tucson	81489 Crowley Avenue	520-769-9489
3	6003	Provider AA	Hospital	Kentucky	Louisville	38713 Kipling Place	502-275-0043
4	6004	Provider AA	Specialist Service	Georgia	Atlanta	37026 Cambridge Drive	678-427-0176
5	6005	Provider AA	Hospital	New Mexico	Albuquerque	4049 Eagle Crest Road	505-199-2147
6	6006	Provider CC	Specialist Service	Florida	Tallahassee	05338 Artisan Avenue	850-604-2898
7	6007	Provider DD	Specialist Service	California	Los Angeles	93398 Rockefeller Drive	323-771-1946
8	6008	Provider AA	Urgent Care	West Virginia	Charleston	2 Forest Dale Avenue	304-187-0755
9	6009	Provider DD	Specialist Service	North Carolina	Raleigh	3 Banding Way	919-688-2596
10	6010	Provider BB	Clinic	Nebraska	Lincoln	102 Fair Oaks Crossing	402-848-7459

```
insert into employee values
(2001,'Tudor','Churchlow',    '1974-06-23', 'Officer',
'2010-07-17', '802-208-2338', 'tchurchlow0@ca.gov', 'Vermont',
'Montpelier', '36561 Manley Road', 6010);
```

	empID	firstName	lastName	dateOfBirth	jobTitle	hiredDate	phoneNumber	email	state	city	address	zip_code	providerID
1	2001	Tudor	Churchlow	1974-06-23	Officer	2010-07-17	802-208-2338	tchurchlow0@ca.gov	Vermont	Montpelier	36561 Manley Road	12457	6010
2	2002	Lezley	Futcher	1987-03-11	Reviewer	2012-07-04	914-489-1607	lfutcher1@europa.eu	New York	White Plains	389 Charing Cross Park	54875	6007
3	2003	Tiertza	Croisier	2002-01-27	Director	2022-09-19	559-922-8679	tcroisier2@tinyurl.com	California	Fresno	7174 Melody Alley	55447	6001
4	2004	Kellia	Wardrope	1978-07-05	Administrator	2011-06-24	202-554-8184	kwardrope3@uiuc.edu	District of Columbia	Washington	0 Killdeer Parkway	66658	6009
5	2005	Jamima	Goody	1997-05-08	Reviewer	2015-12-15	414-590-3100	jgoody4@tiny.cc	Wisconsin	Milwaukee	046 Huxley Center	77854	6007
6	2006	Tirrell	Rangle	1974-10-13	Reviewer	2019-08-27	941-565-2128	trangle5@harvard.edu	Florida	North Port	1 Marquette Circle	10004	6001
7	2007	Isa	Linzey	1997-09-28	Officer	2019-04-27	915-344-7700	ilinzey6@marketwatch.com	Texas	El Paso	74 Westend Crossing	98855	6004
8	2008	Sidonia	Eary	1972-03-04	Officer	2014-09-23	706-920-0858	seary7@stumbleupon.com	Georgia	Cumming	195 Karstens Court	45698	6003
9	2009	Stephi	Stannas	1986-05-06	Receptionist	2012-07-02	206-524-2957	sstannas8@dagondesign.com	Washington	Seattle	76120 Jana Lane	45454	6005
10	2010	Margit	Diack	1970-05-30	Manager	2018-02-15	904-858-8186	mdiack9@thetimes.co.uk	Florida	Jacksonville	391 Stoughton Park	11441	6008

```
insert into claim_approved values
(7001,'ACVF9','920004','appeal','preventative', '2019-06-27',
6007, 4004, 3004, 5002,270, 'approved',  'A001',  '2019-07-17'),
```

	claimID	diagnosisCode	CPTCode	type	serviceProvided	dateOfService	providerID	subscriberID	planID	dependentID	billedAmount	status	approvalID	submissionDate
1	7001	ACVF9	920004	appeal	preventative	2019-06-27	6007	4004	3004	5002	270	approved	A001	2019-07-17
2	7002	ACVF9	965430	new	medication	2022-02-12	6005	4004	3003	5008	536	approved	A002	2022-04-10
3	7003	87YFG	911231	new	diagnostic	2023-04-19	6004	4001	3004	5001	561	approved	A003	2023-04-30
4	7004	87YFG	965430	appeal	preventative	2020-07-24	6010	4006	3007	5005	679	pending	A004	2020-08-15
5	7005	W223T	900765	appeal	emergency	2023-08-29	6009	4009	3007	5007	922	approved	A005	2023-10-10
6	7006	W223T	920004	appeal	preventative	2022-07-28	6004	4008	3004	5005	789	pending	A006	2022-08-17
7	7007	TH48F	911231	new	emergency	2022-04-28	6005	4002	3004	5006	993	pending	A007	2022-06-01
8	7008	W223T	900765	new	diagnostic	2021-07-16	6003	4005	3003	5008	197	approved	A008	2021-07-24
9	7009	KL3E0	900765	new	medication	2020-12-03	6008	4009	3010	5001	57	pending	A009	2020-12-19
10	7010	TH48F	965430	appeal	medication	2019-06-12	6001	4003	3005	5005	919	approved	A010	2019-06-29

```
insert into claim_denied values
(7011, 'P980J', '900234', 'new', 'diagnostic',  '2020-06-10',
6010, 4002, 3005, 5009, 745, 'denied',  'D001',  '2020-06-15');
```

	claimID	diagnosisCode	CPTCode	type	serviceProvided	dateOfService	providerID	subscriberID	planID	dependentID	billedAmount	status	denialID	submissionDate
1	7011	P980J	900234	new	diagnostic	2020-06-10	6010	4002	3005	5009	745	denied	D001	2020-06-15
2	7012	P09JU	908876	new	surgical	2021-04-26	6004	4004	3001	5005	105	pending	D002	2021-05-20
3	7013	P980J	236543	new	emergency	2022-07-11	6004	4003	3002	5010	145	pending	D003	2022-07-19
4	7014	W445R	236543	appeal	preventative	2023-02-21	6006	4009	3002	5003	759	pending	D004	2023-03-04
5	7015	UYTB4	944551	new	medication	2019-07-01	6008	4010	3002	5001	464	denied	D005	2019-07-12
6	7016	VFR6G	799003	new	surgical	2020-02-17	6007	4010	3005	5009	322	denied	D006	2020-02-29
7	7017	UYTB4	900234	appeal	diagnostic	2022-09-21	6006	4006	3005	5004	415	pending	D007	2022-09-28
8	7018	VFR6G	226688	new	emergency	2019-07-11	6006	4005	3001	5003	934	denied	D008	2019-07-15
9	7019	W445R	908876	appeal	diagnostic	2022-11-02	6004	4009	3001	5002	443	denied	D009	2022-11-10
10	7020	P980J	908876	new	surgical	2020-12-24	6002	4010	3008	5001	959	denied	D010	2020-12-30

Alter Table Statements

```
alter table subscriber
add age_group varchar(5)
```

```
 alter table claim_denied
```

```
add denial_reason varchar(100)
```

VII. Scenarios

Scenario 1

How many subscribers are enrolled in each plan type?

- Shows how each plan is performing in terms of subscriber base for marketing purposes.

Query

```
select planType, count(s.subscriberID) as #ofSubscribers
from subscriber s,[plan] p, plan_subscriber ps
where s.subscriberID = ps.subscriberID and p.planID = ps.planID
group by planType
```

Scenario 2

How many subscribers over the age of 50 are enrolled in each plan?

- Subscribers aged 50+ are more likely to have more claims because of more health concerns.

Query

```
select planType,  count(s.subscriberID) as #ofSubscribers
from subscriber s,[plan] p, plan_subscriber ps
where s.subscriberID = ps.subscriberID
     and p.planID = ps.planID
     and datediff(year, dateOfBirth, getdate()) > 50
group by planType
```

Scenario 3

What is the average billed amount for claims under a specific plan?

- Will aid in cost analysis, setting premiums, managing expenses, and negotiating contracts.

Query

```
select planType, round(avg(billedAmount),2) avgBilledAmnt
from claim_approved c, [plan] p
where c.planID = p.planID
group by planType
```

Scenario 4

How many claims are generated each month per provider?
- Will aid in evaluating provider performance. Can also be used to forecast future claims per provider.

Query

```
with claim_data as(
 select providerID, format(submissionDate, 'yyyy-MM')
submissionMonth,
 count(providerID) #ofclaims from [claim_approved]
 group by providerID, format(submissionDate, 'yyyy-MM')
 union all
 select providerID, format(submissionDate, 'yyyy-MM')
submissionMonth,
 count(providerID) #ofclaims from [claim_denied]
 group by providerID, format(submissionDate, 'yyyy-MM')
 )
 select providerID, submissionMonth,
 sum(#ofclaims) total_claims from claim_data
 group by providerID, submissionMonth
```

Scenario 5

What percentage of total premiums collected is paid out in claims?
- This will help in determining the profitability of the health insurance business.

Query

```
WITH AggregatedAmounts AS (
    SELECT
        (SELECT SUM(billedAmount) FROM claim_approved) AS
TotalBilledAmount,
        (SELECT SUM(premiumAmount) FROM [plan]) AS
TotalPremiumAmount
)
SELECT ROUND((TotalBilledAmount / TotalPremiumAmount) * 100, 2)
AS ClaimToPremiumRatio
FROM AggregatedAmounts;
```

Scenario 6

Which type of provider facilities are most utilized by plan subscribers?

- Will help establish the preferred provider choice by type and determine which facility types should be expanded in future operations.

Query

```
select p.facilityType, count(t.subscriberID) total_subs from
[provider] p
inner join
(select subscriberID, providerID from claim_approved
union all
select subscriberID, providerID from claim_denied) t
on p.providerID = t.providerID
group by p.facilityType
```

VIII. Conclusion

The development of HealthHub Systems has been a profoundly enriching experience for our team. For our group, it began with finding a suitable topic for building a database. This was somewhat challenging. After weighing concepts in movies and other industries we eventually settled on the health insurance industry. Another challenge we faced was limited direct experience in the healthcare insurance industry. We successfully navigated through this complexity by leveraging our diverse backgrounds and extensive research. The project enabled us to transition theoretical knowledge into practical application, guiding us through the intricate processes of conceptualization, negotiation, and execution. HealthHub Systems now stands as a testament to our collaborative efforts – a robust database tool tailored to meet the data management needs of the healthcare insurance industry. We believe that HealthHub Systems will serve as a significant asset for the industry, facilitating better decision-making and operational efficiencies through its comprehensive capabilities.